Fish Abundance Over Time

```
# loading in packages
library(tidyverse) # general use
library(janitor) # cleaning data
library(here) # data organization
library(lubridate) # date and time

fish <- read.csv(here("data",
                      "Monthly_Fish_All_Years.csv"))
```

```
# creating new object, fish_clean
fish_clean <- fish |>
  # cleaning the names
  clean_names() |>
  # renaming the date column to original_date
  rename(original_date = date) |>
  # filtering to remove the no count data, -99999
  filter(count >= 0) |>
  # using mutate and lubridate to convert original_date
  # into new date column that can be used to create time series
  mutate(
    date = ymd(original_date),
    year_month = floor_date(date, unit = "month")) |>
  # relocating columns to to left to be easier to look at
  relocate(year_month, count, common_name, site)
```

```
# creating top_species_table
top_species_table <- fish_clean |>
  # grouping by species
  group_by(common_name) |>
  # summarize to make one row per species
  # sum to get the total fish count of each species
```

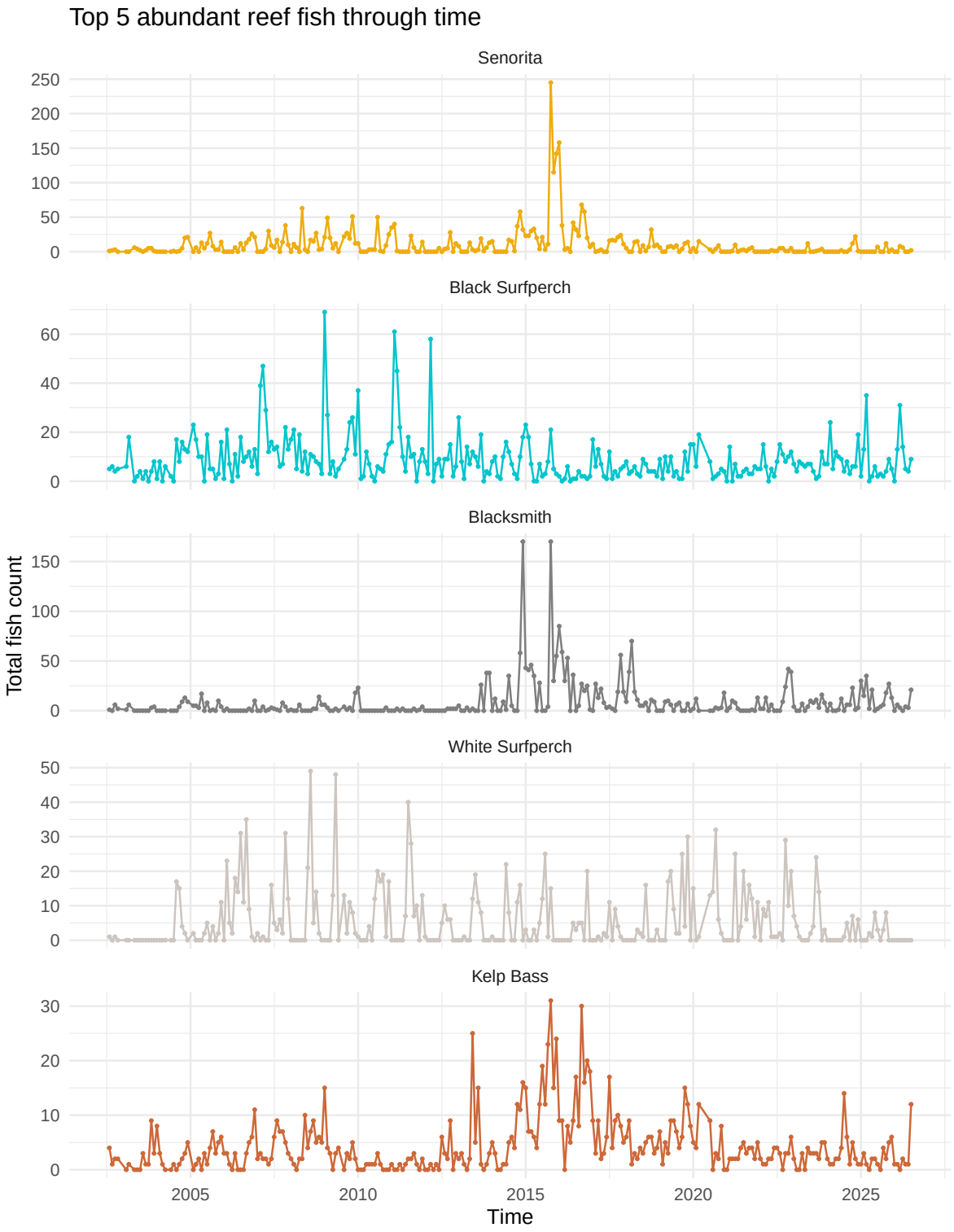
```
  summarize(total_count = sum(count)) |>
  # ungrouping the data
  ungroup() |>
  # ranking from most to least abundant
  arrange(desc(total_count)) |>
  # selecting top 5 species
  slice_head(n = 5)

# creating top_species_vector
top_species_vector <- top_species_table |>
  # using pull to extract column as a vector
  pull(common_name)
```

```
# creating new object fish_ts
fish_ts <- fish_clean |>
  # filtering for the top_species_vector in common_name
  filter(common_name %in% top_species_vector) |>
  # using fct_relevel to change order within common_name
  mutate(common_name = fct_relevel(
    common_name,
    top_species_vector)) |>
  # grouping data by date and species
  group_by(year_month,
           common_name) |>
  # summarize to make one row per month
  # sum to get the total fish count of each species
  summarize(total_count = sum(count)) |>
  # ungrouping the data
  ungroup() |>
  # arranging by most abundant species and by date
  arrange(common_name, year_month)
```

```
# base plot
ggplot(fish_ts,
       # x-axis: time
       aes(x = year_month,
           # y-axis: abundance
           y = total_count,
           # coloring by species
           color = common_name)) +
  # first layer: geom_line
  geom_line() +
```

```
  # second layer: geom_point
  geom_point(size = 0.5) +
  # faceting by species
  facet_wrap(~ common_name,
             # putting into one column
             ncol = 1,
             # making each panel get its own y range
             scales = "free_y") +
  labs(
    # adding a graph title
    title = "Top 5 abundant reef fish through time",
    # naming x axis Time
    x = "Time",
    # naming y axis Total fish count
    y = "Total fish count") +
  # customizing colors
  scale_color_manual(values = c(
    "Senorita" = "darkgoldenrod2",
    "Black Surfperch" = "turquoise3",
    "Blackfish" = "slategray",
    "White Surfperch" = "seashell3",
    "Kelp Bass" = "sienna3")) +
  # adding custom theme
  theme_minimal() +
  # removing legend
  theme(legend.position = "NONE")
```


Top 5 abundant reef fish through time
Senorita
Black Surfperch
Blacksmith
White Surfperch
Kelp Bass
Total fish count
Time
2005
2010
2015
2020
2025